An Angle-Dependent Impedance Boundary Condition for the Split-Step Parabolic Equation Method

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***Abstract*—A formulation of the impedance boundary condition (IBC) is derived, which can be used in split-step parabolic wave equation solvers to accurately represent an interface with arbitrary dielectric properties. The approach relies on ensuring that the plane-wave decomposition of the field satisfies the appropriate IBC for each spectral component. Numerical experiments illustrating the robustness of the approach for low-contrast interfaces and angles-of-incidence near the Brewster angle are presented.**

***Index Terms*—Electromagnetic propagation, Maxwell equations, microwave propagation.**

I. Introduction

PARABOLIC equation solvers [1], [2] provide a powerful modeling capability for propagation of electromagnetic waves over long distances in complex environments. They are used extensively for predicting radar coverage in ducting enviroments over rough surfaces. Two primary classes of solvers are used in conjuction with the parabolic wave equation. The first is a finite-difference approach that relies on fine discretization of the spatial domain for accurate representation of the field propagation. The second is the split-step approach [3], [4] that implements propagation in the Fourier domain and utilizes a series of "phase-screens" to account for refraction effects.

One of the primary advantages of the split-step method with respect to the finite-difference approach is its insensitivity to discretization size, i.e., while finite-difference methods may require discretization in both altitude and range on the order of a tenth of a wavelength or finer, the split-step method needs only half-wavelength discretization in altitude while range steps can be large compared to the wavelength. In practice, the discretization for the split-step method is driven by the variation in the environment rather than a requirement to resolve the fast variation of the fields.

A consequence of the use of Fourier techniques, however, is that the split-step method requires that the dielectric properties of the domain be slowly varying in both altitude and range. By contrast, the finite-difference method is able to handle sharp interfaces between regions of different dielectric properties naturally by solving for the fields both above and below the interface and enforcing continuity of the tangential fields at the interface. In the split-step method, propagation over such an interface must be treated as propagation in a half-space along with an appropriate boundary condition at the interface.

In this paper, we present a new method for treating the interface using an impedance boundary condition (IBC). The investigation leading to this result was motivated by the problem of propagation over lunar terrain that has near-zero conductivity and low permittivity ($\epsilon_{\mathrm{r}} \approx 2$). Previous implementations of the IBC did not handle such low-constrast interfaces accurately, and so a new formulation was required. The resulting "consistent-field" IBC formulation is presented in Section II with numerical experiments for the lunar terrain in Section III.

II. Formulation

A. Split-Step Method

In this paper, we consider a cylindrical coordinate system with azimuthal symmetry in both the fields and environment. The Helmholtz equation for this geometry is

$$\left(\partial_r^2 + \partial_z^2 + k_0^2 n^2 + r^{-2}/4\right) u = 0 \tag{1}$$

where $u(r,z)$ is the auxiliary field which is equal to $E_\phi \sqrt{k_0 r}$ for horizontal polarization (H-pol) and $H_\phi \sqrt{k_0 r}$ for vertical polarization (V-pol). Here and throughout this paper, we assume and omit a harmonic time variation of $e^{-i\omega t}$ for all fields. In the far field, the term proportional to r^{-2} may be neglected. Factorizing the resulting far-field wave equation into forward and backward components gives

$$(\partial_r + iQ_n)(\partial_r - iQ_n)u = i[Q_n, \partial_r]u \tag{2}$$

where

$$Q_n \equiv \sqrt{k_0^2 n^2 + \partial_z^2}. \tag{3}$$

Assuming that the commutator $[Q_n, \partial_r]$ is negligible and considering only outgoing waves, we have the *parabolic equation*

$$\partial_r u = iQ_n u. \tag{4}$$

This equation has the formal solution

$$u(r + \Delta r, z) = e^{iQ_n \Delta r} u(r,z). \tag{5}$$

The well-known (wide-angle) split-step method computes the field taking into account the refractivity by making the

Manuscript received July 29, 2010; revised June 27, 2011; accepted August 22, 2011. Date of publication October 21, 2011; date of current version February 03, 2012.

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Digital Object Identifier 10.1109/TAP.2011.2173107

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approximation (see [4], for example; this approximation was introduced and discussed in [5])

$$Q_n = \sqrt{k_0^2 n^2 + \partial_z^2} \approx k_0(n-1) + Q_1 \tag{6}$$

with $Q_1 \equiv \sqrt{k_0^2 + \partial_z^2}$ from (3). With this approximation, (5) is approximated as

$$u(r+\Delta r, z) \approx e^{ik_0 \frac{n-1}{2}\Delta r} e^{iQ_1 \Delta r} e^{ik_0 \frac{n-1}{2}\Delta r} u(r,z) \tag{7}$$

where the symmetric splitting of the "refractive" portion of the propagator is used and $n = n(r,z)$ is allowed to be a slowly varying function of both range and altitude. The "diffractive" portion of the propagator, $\exp(iQ_1\Delta r)$, is handled in the Fourier p-space conjugate to the z-coordinate. In p-space, the diffractive propagator is thus $\exp(iq\Delta r)$, where $q \equiv \sqrt{k_0^2 - p^2}$. If, in addition to the slow variation assumption required for the above derivation, we have $n^2(r,z) - 1 \ll 1$, then this propagator is accurate for angles $\theta = \arctan(p/q)$ from -90° to $+90^\circ$. In particular, it is exact for free-space propagation.

The split-step procedure for advancing the field thus becomes

$$u(r+\Delta r, z) \approx e^{ik_0 \frac{n-1}{2}\Delta r} \times \mathcal{F}^{-1}\left\{ e^{iq\Delta r} \mathcal{F}\left[e^{ik_0 \frac{n-1}{2}\Delta r} u(r,z) \right] \right\} \tag{8}$$

where $\mathcal{F}$ represents the Fourier transform with respect to the z-coordinate.

B. Impedance Boundary Condition

Propagation above a discontinuous dielectric interface using the split-step method must be incorporated in the diffractive propagator due to the slow-variation requirement for $n(r,z)$. This interface is incorporated by considering only fields in the half-space bounded below by the interface and using an impedance boundary condition to include the effects of the presence of the interface. While the full boundary condition at the interface requires continuity of both tangential electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$, at the boundary, the half-space problem admits only a single boundary condition. Thus, a single impedance condition on the propagated field given by

$$(\partial_z u + i\alpha u)|_{z=0} = 0 \tag{9}$$

is used. This condition is valid for the case where the field $u(r,z)$ is a plane wave of the form

$$u(r,z) = u_0 e^{i(qr+pz)} \tag{10}$$

with $p^2 + q^2 = k_0^2$ (we take $p < 0$ for waves having a component that propagates in the negative z-direction). For V-pol, the propagated field is H_ϕ and from Ampére's Law, we have $\partial_z H_\phi = i\omega\epsilon E_r$. Denoting the fields below the surface with primes, a linear combination of the boundary conditions on the tangential $\mathbf{E}$ and $\mathbf{H}$ fields is

$$(E_r + \beta H_\phi)|_{z=0} = (E_r' + \beta H_\phi')|_{z=0}. \tag{11}$$

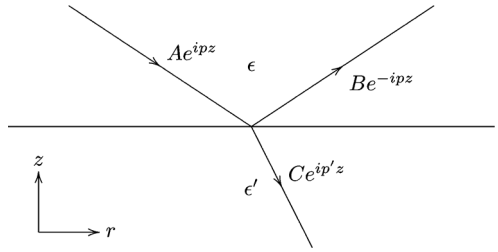


Fig. 1. Plane wave incident on the interface between two homogeneous half-spaces. Note that with the sign convention used here, $p, p' < 0$ in this figure.

Elimination of the subsurface fields from the boundary condition requires that

$$\beta = -\frac{E_r'}{H_\phi'} = -\frac{1}{i\omega\epsilon'}\frac{\partial_z H_\phi'}{H_\phi'}. \tag{12}$$

Considering fields of the form (10), the incident and reflected fields are given by

$$H_\phi^{\mathrm{i}} = \frac{A}{\sqrt{k_0 r}} e^{i(qr+pz)} \tag{13}$$

$$H_\phi^{\mathrm{r}} = \frac{B}{\sqrt{k_0 r}} e^{i(qr-pz)} \tag{14}$$

respectively in the upper domain (see Fig. 1), while the field in the lower domain is given by

$$H_\phi' = \frac{C}{\sqrt{k_0 r}} e^{i(qr+p'z)} \tag{15}$$

where $p' = -\sqrt{(\epsilon_{\mathrm{r}} - 1)k_0^2 + p^2}$ with $\epsilon_{\mathrm{r}} = \epsilon'/\epsilon$. Hence, β is given by

$$\beta = -\frac{p'}{\omega\epsilon'}. \tag{16}$$

Substituting (16) into (11) gives

$$\left(\partial_z H_\phi - i\frac{p'}{\epsilon_{\mathrm{r}}} H_\phi \right)\Big|_{z=0} = 0. \tag{17}$$

The H-pol case is similar, providing the proper definition of the coefficient α in (9) as

$$\alpha(p) = \begin{cases} \sqrt{(\epsilon_{\mathrm{r}} - 1)k_0^2 + p^2}/\epsilon_{\mathrm{r}}, & \text{V-pol} \\ \sqrt{(\epsilon_{\mathrm{r}} - 1)k_0^2 + p^2}, & \text{H-pol} \end{cases} \tag{18}$$

where the branch cut of the square-root is taken to be along the negative real axis such that $\Re(\alpha) \geq 0$. The boundary condition for the field in the upper domain then gives us the condition

$$(p+\alpha)A = (p-\alpha)B \tag{19}$$

for the field amplitudes. The traditional IBC formulation [6] assumes a unit incident field amplitude, $A = 1$, such that $B = \Gamma(-p)$, where $\Gamma(p)$ is the reflection coefficient

$$\Gamma(p) = \frac{p-\alpha}{p+\alpha}. \tag{20}$$

It should be noted that p is associated with the direction of propagation of the *incident* field, which explains the choice of sign in the definition of B. With this formulation, the field $u(r+\Delta r, z)$ in terms of the field $u(r, z)$ is given by

$$u(r+\Delta r, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{u}(r,p) e^{i(pz+q\Delta r)}\, dp \tag{21}$$

where

$$\hat{u}(r,p) \equiv \int_0^{\infty} u(r,z)[e^{-ipz} + \Gamma(p) e^{ipz}]\, dz. \tag{22}$$

Here and in the remainder of this paper, we neglect any surface wave contributions and focus solely on the spatial waves. For the case of lossless dielectrics (real α) considered in Section III, there is no surface wave component, and the results presented are exact.

Equation (22) can be written in terms of the regular Fourier transforms of the even (u_+) and odd (u_-) extensions of u as

$$\hat{u}(r,p) = \frac{1+\Gamma(p)}{2}\tilde{u}_+(r,p) + \frac{1-\Gamma(p)}{2}\tilde{u}_-(r,p) \tag{23}$$

where

$$\tilde{f}(r,p) = \mathcal{F}[f(r,z)] \equiv \int_{-\infty}^{\infty} f(r,z) e^{-ipz}\, dz. \tag{24}$$

Equation (21) then becomes

$$u(r+\Delta r, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{p\tilde{u}_+ + \alpha\tilde{u}_-}{p+\alpha} e^{ipz}\, dp \tag{25}$$

where $\tilde{u}_\pm = \tilde{u}_\pm(r+\Delta r, p) = \tilde{u}_\pm(r,p)\exp(iq\Delta r)$. This is the "mixed-transform" approach. The singularity at $\alpha(p) = -p$ in the integrand may be handled by taking the Cauchy Principal Value (CPV) of the integral. As will be shown in Section III, this formulation exhibits artifacts associated with this singularity. Alternative handling of the singular integrand does not resolve the issue. For example, adding and subtracting the singular part of the integrand to construct a regular integrand that can be numerically integrated and analytically integrating the singular part gives

$$u(r+\Delta r, z) \equiv u_{\mathrm{r}}(r+\Delta r, z) + u_{\mathrm{s}}(r+\Delta r, z) \tag{26}$$

with

$$u_{\mathrm{r}}(r,z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left[\frac{p\tilde{u}_+ + \alpha\tilde{u}_-}{p+\alpha} - \frac{R(r)}{p-p_*}\right] e^{ipz}\, dp \tag{27}$$

where $R(r)$ is the residue at $p_* = p_{\mathrm{r}} + ip_{\mathrm{i}}$, and

$$u_{\mathrm{s}}(r,z) = \frac{i}{2} R(r) e^{ip_* z}\mathrm{sign}z. \tag{28}$$

Here, we have used the relation

$$\int_{-\infty}^{\infty} \frac{e^{ipz}}{p}\, dp = i\pi \mathrm{sign}z. \tag{29}$$

For $p_{\mathrm{i}} = 0$, the residue is given by

$$R(r) = \frac{1}{1+\alpha_{\mathrm{r}}'}[p_{\mathrm{r}}\tilde{u}_+(r,p_{\mathrm{r}}) + \alpha_{\mathrm{r}}\tilde{u}_-(r,p_{\mathrm{r}})] \tag{30}$$

where $\alpha_{\mathrm{r}} \equiv \alpha(p_{\mathrm{r}})$ and $\alpha_{\mathrm{r}}' \equiv \alpha'(p_{\mathrm{r}})$, while for $p_{\mathrm{i}} \neq 0$, we set

$$\begin{aligned} R(r) &= \lim_{p\to p_{\mathrm{r}}} (p-p_*)\frac{p\tilde{u}_+ + \alpha\tilde{u}_-}{p+\alpha} \\ &= \frac{-ip_{\mathrm{i}}}{p_{\mathrm{r}}+\alpha_{\mathrm{r}}}\left[p_{\mathrm{r}}\tilde{u}_+(r,p_{\mathrm{r}}) + \alpha_{\mathrm{r}}\tilde{u}_-(r,p_{\mathrm{r}})\right]. \end{aligned} \tag{31}$$

This method exhibits numerical artifacts similar to those encountered with the simpler CPV approach.

Previous efforts [7], [8] have sought to mitigate these artifacts through the use of a variety of numerical techniques, but rely on an angle-independent impedance assumption, $\alpha(p) \equiv \alpha_0$, such as $\alpha(p) \equiv \alpha(0)$ [9]. One such approach defines an auxiliary field

$$w(z) \equiv (\partial_z + i\alpha_0)u(z) \tag{32}$$

and solves for $w(z)$ using sine transforms (the boundary condition for $w(z)$ is simply $w(0) = 0$). The field $u(z)$ is then computed as the solution to the forced differential equation (32). If $w(z)$ has any components proportional to the homogeneous solution to this equation, $\exp(-i\alpha_0 z)$, the solution for $u(z)$ will diverge due to resonance. The alternative central/forward/backward difference approach of [8] effectively shifts the homogeneous solution away from any component of $w(z)$ resulting in a well-behaved solution. However, this approach has the undesirable side-effect of changing the reflection amplitude for incidence angles away from grazing. An example of this effect is shown in Section III.

In contrast, this paperrevisits (19) to derive a formulation that avoids the singular behavior of (25).

C. Consistent-Field IBC Formulation

For a general field, (19) must be satisfied for the plane-wave decomposition of the field $u(z)$, i.e.,

$$(p+\alpha)\tilde{u}(p) = (p-\alpha)\tilde{u}(-p). \tag{33}$$

This formulation focuses on ensuring that the propagated field is consistent with the presence of the interface. In particular, $\tilde{u}(0) = 0$ and $\tilde{u}(p_0) = 0$, where p_0 is the solution to $p = \alpha(p)$. For H-pol, there is no such solution, while for V-pol, there is a solution, $p_0 = p_{\mathrm{B}}$, which corresponds to boundary reflections of waves incident on the surface at the Brewster angle.

As the boundary condition is entirely accounted for through modification of the diffractive part of (8), without loss of generality, we consider the case with $n = 1$, such that

$$u(r+\Delta r, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{u}(r,p) e^{i(pz+q\Delta r)}\, dp \tag{34}$$

where $\tilde{u}(r,p)$ must satisfy (33). By decomposing $u(z)$ into the sum of functions $w(z)$ and $v(z)$, which are respectively even and odd in z, and substituting into (33), it is shown that

$$\tilde{u}(p) = \left(1 - \frac{p}{\alpha}\right)\tilde{v}(p) \tag{35}$$

where $\tilde{v}(p) = -\tilde{v}(-p)$. Here, we have used the fact that the Fourier transform preserves even/odd symmetry, i.e.,

$$f(z) = \pm f(-z) \Longrightarrow \tilde{f}(p) = \pm \tilde{f}(-p). \tag{36}$$

Thus, the problem may be solved completely in terms of an odd function of z. The intial field $u(z)$ is specified for $z \geq 0$, from which we must determine $v(z) = -v(-z)$ such that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} dp\, e^{ipz} \left(1 - \frac{p}{\alpha}\right) \times \int_{-\infty}^{\infty} v(z') e^{-ipz'}\, dz' = u(z) \qquad \forall z \geq 0. \tag{37}$$

To this end, define

$$a(z) \equiv \begin{cases} u(z), & z \geq 0 \\ 0, & z < 0 \end{cases} \tag{38}$$

$$b(z) \equiv \begin{cases} 0, & z \geq 0 \\ u(z), & z < 0 \end{cases} \tag{39}$$

so that $u(z) = a(z) + b(z)$ with $a(z)$ known and $b(z)$ unknown. Taking the Fourier transform, we have

$$\tilde{u}(p) = \tilde{a}(p) + \tilde{b}(p). \tag{40}$$

To solve for $b(z)$, we will consider the discrete Fourier transform such that (40) becomes

$$\tilde{u}_n \equiv \sum_{m=-M+1}^{M} u_m e^{-i\pi nm/M} = \tilde{a}_n + \sum_{m=1}^{M-1} b_{-m} e^{i\pi nm/M} \tag{41}$$

where $b_m = 0$ for $m = 0, \ldots, M$. Substitution into the discrete version of (33) gives

$$\sum_{m=1}^{M-1} \left[\alpha_n \cos\left(\frac{\pi nm}{M}\right) + ip_n \sin\left(\frac{\pi nm}{M}\right)\right] b_{-m} = \frac{p_n - \alpha_n}{2} \tilde{a}_{-n} - \frac{p_n + \alpha_n}{2} \tilde{a}_n \tag{42}$$

which may be solved in the least-squares sense as an overdetermined linear system. This procedure is computationally intensive, but would only need to be performed for the initial field (e.g., at $r = 0$). In practice, for initial fields that are nonzero only away from the interface, it is a good approximation to simply set $u(z) = 0$ for $z \leq 0$. It is this latter approach that is used in the results presented in Section III.

In theory, the field $\tilde{v}(p)$ could simply be computed from (35). For this to be valid, however, it must be true that

$$\left| \lim_{p \to p_0} \frac{\tilde{u}(p)}{\alpha - p} \right| < \infty \tag{43}$$

where, as above, p_0 is the solution to $\alpha(p) = p$. For a general antenna pattern, this condition will not be satisfied. This issue arises from the fact that (33) was derived for a plane-wave decomposition of the fields in which the components exist everywhere above the interface. Thus, any field component that is upwardly propagating must have been reflected from the surface at some point. For $p = p_0$, since $\Gamma(p_0) = 0$, this implies that this component must be zero. Realistically, however, the source is the aperture field of an antenna located at a finite range. The antenna may produce components in the direction p_0, and these must be accurately captured if this method is to be useful in practice. We note that since $\alpha > 0$, $p_0 > 0$ and hence components for which $p \approx p_0$ will not interact with the surface during propagation. Thus, it is a good approximation to propagate these components in free space.

To accomplish this, the incident field can be partitioned into two portions, $\tilde{u}(r,p) = \tilde{u}_{\rm s}(r,p) + \tilde{u}_{\rm f}(r,p)$, propagated in the presence of the surface and in free space, respectively. One such splitting is given by

$$\tilde{u}_{\rm s}(r,p) = \frac{\alpha - p}{\alpha - (p + ik_0\delta)} \tilde{u}(r,p) \tag{44}$$

$$\tilde{u}_{\rm f}(r,p) = \frac{-ik_0\delta}{\alpha - (p + ik_0\delta)} \tilde{u}(r,p) \tag{45}$$

where δ is a unitless parameter that determines the splitting, with larger δ including more components in the free-space field. The field $\tilde{u}_{\rm s}(p)$ is propagated in the presence of the surface by propagating the antisymmetric field $\tilde{v}_{\rm s}(p)$ given by (35), while the field $\tilde{u}_{\rm f}(p)$ is propagated in free space using (8). The total field at the next range step is then given by

$$\tilde{u}(r + \Delta r, p) = \left(1 - \frac{p}{\alpha}\right) \tilde{v}_{\rm s}(r + \Delta r, p) + \tilde{u}_{\rm f}(r + \Delta r, p). \tag{46}$$

A detailed outline of this algorithm is included in the Appendix.

III. Numerical Experiments

Current methods perform well for high-contrast interfaces and/or shallow grazing angles, so the experiments presented here will focus on scenarios in which these methods break down. As the singularity in (25) occurs at the negative of the Brewster angle, a worst-case scenario is expected for excitations near this angle. Therefore, a low-contrast interface at $z = 0$ separating regions of free space (above) and $\epsilon_{\rm r} = 2$ (below) is used. The simulation is initiated with a Gaussian source field [2]

$$u(0,z) = \sqrt{k_0} \tan\theta_{\rm bw} e^{-\frac{1}{2}k_0^2 (z-h)^2 \tan^2\theta_{\rm bw} + ik_0(z-h)\sin\theta_{\rm el}} \tag{47}$$

with $k_0 \approx 15\pi$ m^{-1} ($f = 2.25$ GHz), $h = 25$ m, $\theta_{\rm bw} = 1^\circ$, and $\theta_{\rm el} = -35^\circ$. The value of $\theta_{\rm el}$ was chosen such that the beam is incident on the interface near the Brewster angle $\theta_{\rm B} = \arcsin(1/\sqrt{\epsilon_{\rm r} + 1}) \approx 35.3^\circ$.

Fig. 2 shows the power distribution of the field for this scenario as computed by the mixed-transform method. Severe numerical artifacts dominate the result with variations in the implementation details determining the particular form of the artifacts. Treating the singularity as in (26) yields a similar result. Mitigation techniques such as those described in [8] remove the singular behavior by effectively modifying the surface properties near the Brewster angle.

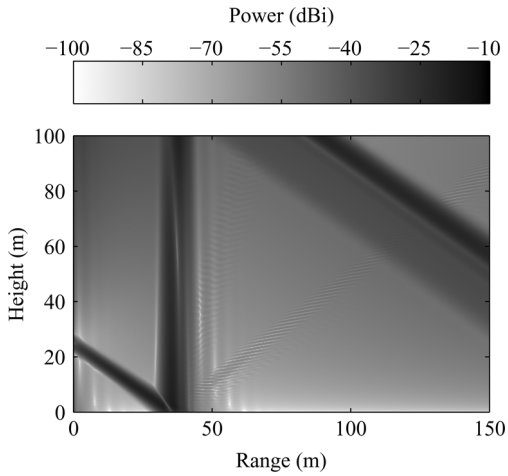


Fig. 2. Gaussian beam with 1° half-beamwidth and $-35°$ elevation angle incident on the interface at $z = 0$ between regions of free space ($z > 0$) and $\epsilon_r = 2 (z < 0)$ computed using the mixed-transform approach. Numerical artifacts dominate the result.

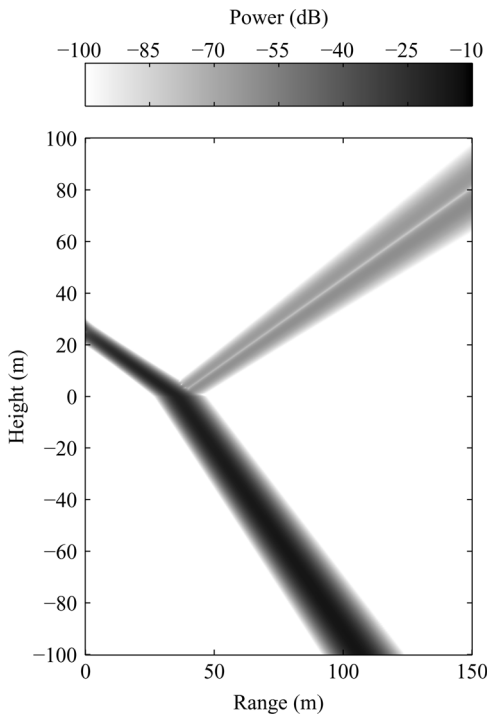


Fig. 3. Gaussian beam with 1° half-beamwidth and $-35°$ elevation angle incident on the interface at $z = 0$ between regions of free space ($z > 0$) and $\epsilon_r = 2 (z < 0)$ computed using finite-differences in both upper and lower half-spaces. Spatial sampling in both height and range is $3\lambda/800$. The null in the reflected field at the Brewster angle is clearly indicated.

The finite-difference solution to the above scenario is shown in Fig. 3. This method solves for the fields in both upper and lower half-spaces and treats the boundary condition at the interface by enforcing continuity of the tangential fields. Hence, there is no impedance boundary condition assumption so that this solution serves as a useful "ground-truth" for comparison with the split-step solutions. Unlike the split-step method, however, the finite-difference method is sensitive to the resolution used. A very fine sampling of $3\lambda/800$ in both range and altitude was used to ensure an accurate baseline for comparisons.

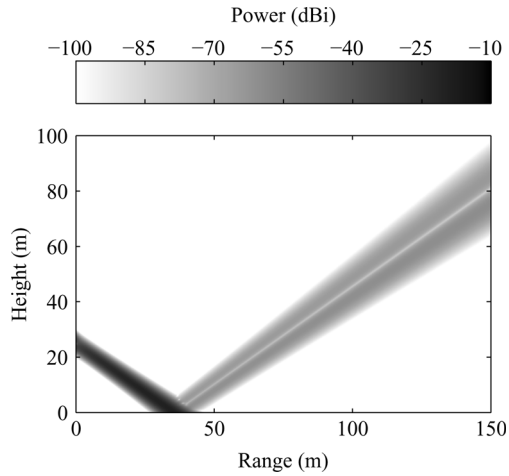


Fig. 4. Gaussian beam with 1° half-beamwidth and $-35°$ elevation angle incident on the interface at $z = 0$ between regions of free space ($z > 0$) and $\epsilon_r = 2 (z < 0)$ computed using the consistent-field formulation. Vertical sampling is $3\lambda/8$. There is a clear null in the reflected power exactly at the Brewster angle $\theta_B = \arcsin(1/\sqrt{\epsilon_r + 1})$ as expected.

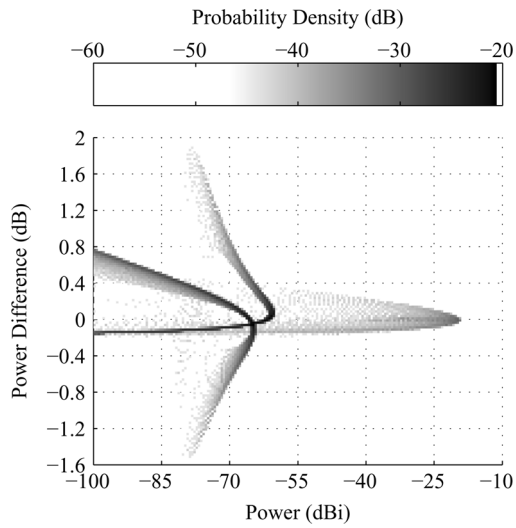


Fig. 5. Histogram of the difference between the results shown in Fig. 4 and the finite-difference result using a spatial resolution of $3\lambda/400$ plotted as a function of power.

Fig. 4 shows the result computed using the consistent-field IBC formulation. The numerical artifacts seen in Fig. 2 are absent, and the expected null in the reflected field along the Brewster angle $\theta_B = \arcsin(1/\sqrt{\epsilon_r + 1})$ is accurately predicted.

Fig. 5 shows a bivariate histogram of the difference between the consistent-field split-step result and the finite-difference result as a function of power over the domain shown in Fig. 4. This plot shows the degree of discrepancy between the two results as well as the power levels at which the differences occur. Large differences at low power levels are less significant than at higher power levels. Agreement between the two results is extremely good with generally less than 1 dB of difference and less than 0.5 dB for power levels greater than -70 dB.

To test the convergence properties of both the split-step and finite-difference methods, the spatial sampling was increased and the simulations rerun. Increasing the altitude sampling of the split-step method from $3\lambda/8$ to $3\lambda/40$ did not affect the

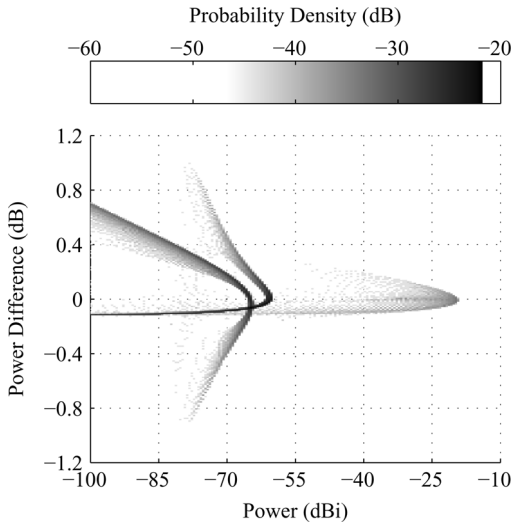


Fig. 6. Histogram of the difference between the results shown in Figs. 4 and 3 as a function of power.

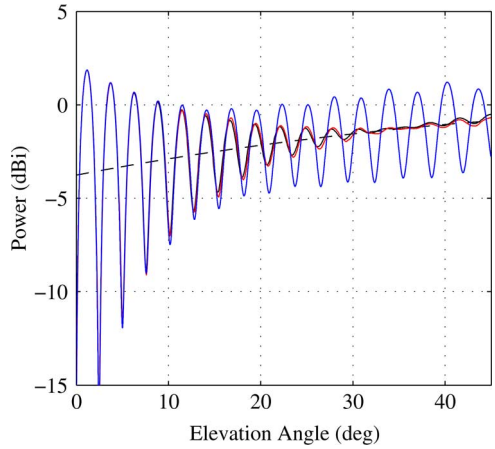


Fig. 7. Tapered cardioid antenna pattern propagated in free space (dashed curve) and over an interface at $z = 0$ between regions of free space ($z > 0$) and $\epsilon_r = 2 (z < 0)$ computed using the finite-difference method (solid black curve), an implementation of the method of [8] (fine dashed curve, blue online), and (46) (dotted curve, red online).

result. Sampling in range was not varied as the split-step method is exact with respect to range for a free-space upper medium as it is simply a process of serial application of Huygens' Principle. Doubling the sampling (in both range and altitude) of the finite-difference method from $3\lambda/400$ to $3\lambda/800$ reduced the error as shown in Fig. 6. Thus, these plots represent a measure of the accuracy of the finite-difference method with the split-step result representing "ground-truth."

An example from the lunar propagation scenario which motivated this investigation is shown in Fig. 7. It shows a comparison of the consistent-field approach to the method of [8] for a wide-angle antenna pattern. The antenna pattern used is the tapered cardioid defined by

$$P(\theta_{\rm el}) = w(\theta_{\rm el}) \left(\frac{1 + \sin \theta_{\rm el}}{2} \right)^{5/8} \tag{48}$$

where $\theta_{\rm el}$ is the elevation angle and w is the taper function

$$w(\theta_{\rm el}) = \begin{cases} \sin^2 2\theta_{\rm el}, & \theta_{\rm el} > \frac{\pi}{4} \\ 1, & \text{otherwise.} \end{cases} \tag{49}$$

This was one of two proposed antenna patterns for the communications system on a lunar lander intended for the International Lunar Network mission currently under study by NASA. The simulation was again performed at a frequency of 2.25 GHz with the antenna located 1.5 m above the interface ($\epsilon_r = 2$). The consistent-field result uses (46) with $\delta = 0.1$ and provides much better agreement with the finite-difference result than the method of [8], which was developed for high-conductivity surfaces and does not correctly capture the Brewster-angle effect (manifested here as a damping of the interference pattern due to the vanishing of the reflected field at the Brewster angle).

IV. Conclusion

A formulation of the impedance boundary condition has been developed, which eliminates divergences seen in previous approaches while preserving the correct behavior of reflections from the surface. The technique is based on an application of the IBC for a single plane wave to the spectral decomposition of a general field. Experiments show excellent agreement with full-space solutions.

It should be noted that the approach described here is an alternate method for representing the impedance boundary condition within the split-step formalism. Although the results presented in this paper are for the case of free space above the surface, atmospheric refractivity can be incorporated *via* (8) in the usual way. Future work will investigate the possibility of including surface-wave effects and combining this approach with techniques for incorporating surface roughness such as the linear shift map [10].

Appendix
Consistent-Field Split-Step Algorithm

This appendix provides an outline of the split-step algorithm using the consistent-field impedance boundary condition incorporating the field-splitting (46).

Let the domain encompass ranges from r_0 to r_M and altitudes from z_0 to z_K with $z_k = k\Delta z$, where the surface is located at $z = 0$. We discretize this domain in range and altitude with steps $\{(r_m, z_k) \mid m = 0, \ldots, M, \text{ and } k = 0, \ldots, K\}$. We image the domain in altitude so that we have $z_{-k} = -z_k$ for $k = 1, \ldots, K$. The algorithm to propagate the auxilliary field $u_{m,k} = u(r_m, z_k)$ to the next range step $u_{m+1,k}$ consists of the following steps.

1) Multiply the field above the surface by the first half of the refractivity correction for the range step

$$u^0_{m,k} = e^{ik_0 \frac{n_{m,k}-1}{2}(r_{m+1}-r_m)} u_{m,k} \tag{50}$$

for $k = 0, \ldots, K$.

2) Apply a z-space windowing function to simulate an unbounded upper half-space by forcing the field to zero at the z_K boundary

$$u^1_{m,k} = w(z_k) u^0_{m,k} \tag{51}$$

for $k = -K, \ldots, K$. Many choices of filter will serve this purpose; we have used a tapered cosine window

$$w(z) \equiv \begin{cases} 1, & |z| \le \frac{4}{5} z_K \\ \frac{1}{2} + \frac{1}{2}\cos\left(5\pi \frac{z}{z_K} - 4\pi\right), & |z| > \frac{4}{5} z_K. \end{cases} \quad (52)$$

3) Transform the field to p-space

$$\tilde{u}^1_{m,j} = \sum_{k=-K}^{K} u^1_{m,k} e^{-i\frac{2\pi jk}{2K+1}} \quad (53)$$

for $j = -K, \ldots, K$.

4) Split the field into the two portions needed to propagate to the next range step and compute the v-field

$$\tilde{v}^{1\mathrm{s}}_{m,j} = \frac{\alpha_j}{\alpha_j - (p_j + ik_0\delta)} \tilde{u}^1_{m,j} \quad (54)$$

$$\tilde{u}^{1\mathrm{f}}_{m,j} = \frac{ik_0\delta}{\alpha_j - (p_j + ik_0\delta)} \tilde{u}^1_{m,j}. \quad (55)$$

5) Transform both fields back to z-space

$$v^{1\mathrm{s}}_{m,k} = \frac{1}{2K+1} \sum_{j=-K}^{K} \tilde{v}^{1\mathrm{s}}_{m,j} e^{i\frac{2\pi jk}{2K+1}} \quad (56)$$

$$u^{1\mathrm{f}}_{m,k} = \frac{1}{2K+1} \sum_{j=-K}^{K} \tilde{u}^{1\mathrm{f}}_{m,j} e^{i\frac{2\pi jk}{2K+1}}. \quad (57)$$

6) Enforce antisymmetry on the v^{s}-field to simulate the effect of the surface and taper the u^{f}-field to zero below the surface

$$v^{2\mathrm{s}}_{m,k} = v^{1\mathrm{s}}_{m,|k|} \mathrm{sign} k \quad (58)$$

$$u^{2\mathrm{f}}_{m,k} = w_{\mathrm{t}}(z_k) u^{1\mathrm{f}}_{m,k} \quad (59)$$

for $k = -K, \ldots, K$ where

$$w_{\mathrm{t}}(z) \equiv \begin{cases} 1, & 0 \le z \\ \frac{1}{2} + \frac{1}{2}\cos\left(\pi \frac{z}{h}\right), & 0 > z > -h \\ 0, & z \le -h \end{cases} \quad (60)$$

for some positive h (we used $h = 50\Delta z$ for the results presented above).

7) Transform the fields to p-space and propagate to the next range step

$$\tilde{v}^{2\mathrm{s}}_{m+1,j} = \tilde{v}^{2\mathrm{s}}_{m,j} e^{iq_j(r_{m+1} - r_m)} \quad (61)$$

$$\tilde{u}^{2\mathrm{f}}_{m+1,j} = \tilde{u}^{2\mathrm{f}}_{m,j} e^{iq_j(r_{m+1} - r_m)}. \quad (62)$$

8) Recombine the fields at the new range

$$\tilde{u}^3_{m+1,j} = \left(1 - \frac{p_j}{\alpha_j}\right) \tilde{v}^{2\mathrm{s}}_{m+1,j} + \tilde{u}^{2\mathrm{f}}_{m+1,j}. \quad (63)$$

9) Apply a p-space window analogous to Step 2.

10) Transform the field back to z-space

$$u^3_{m+1,k} = \frac{1}{2K+1} \sum_{j=-K}^{K} \tilde{u}^3_{m+1,j} e^{i\frac{2\pi jk}{2K+1}}. \quad (64)$$

11) Apply the second half of the refractivity correction

$$u_{m+1,k} = e^{ik_0 \frac{n_{m,k}-1}{2}(r_{m+1} - r_m)} u^3_{m+1,k} \quad (65)$$

for $k = 0, \ldots, K$.

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